Paper Study

High-temperature superconductivity in sulfur hydride evidenced by alternating-current magnetic susceptibility

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PHYSICS

High-temperature superconductivity in sulfur hydride evidenced by alternating-current magnetic susceptibility

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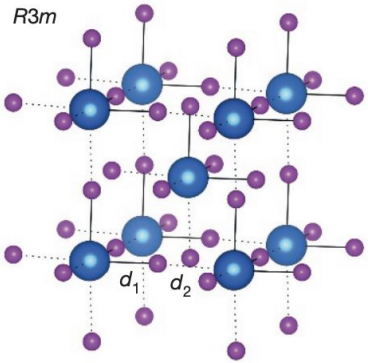
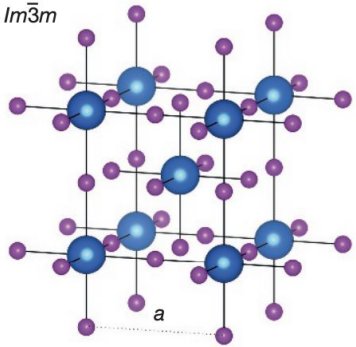
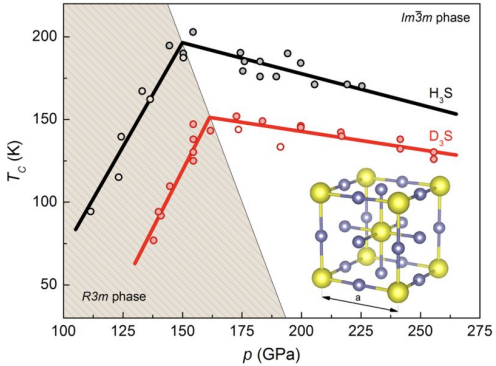
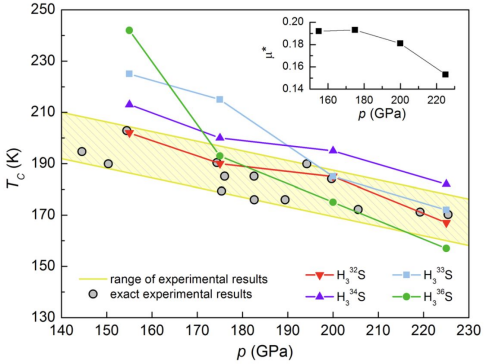
Relevant literature data and underlying principles

- H_3S reaches a maximum Tc of ~203 K at ~155 GPa.
- In the **Im3m structure**, **quantum symmetrization** reduces the pressure required for hydrogen-bond symmetrization, and this phase dominates **the pressure range where high Tc is observed**.
- Replacing sulfur with the heavier isotope **Sulfur-36 is predicted to raise Tc to ~242 K**, and this **anomalous isotope effect** indicates strong phonon coupling, supporting H_3S as a **phonon-mediated superconductor.**



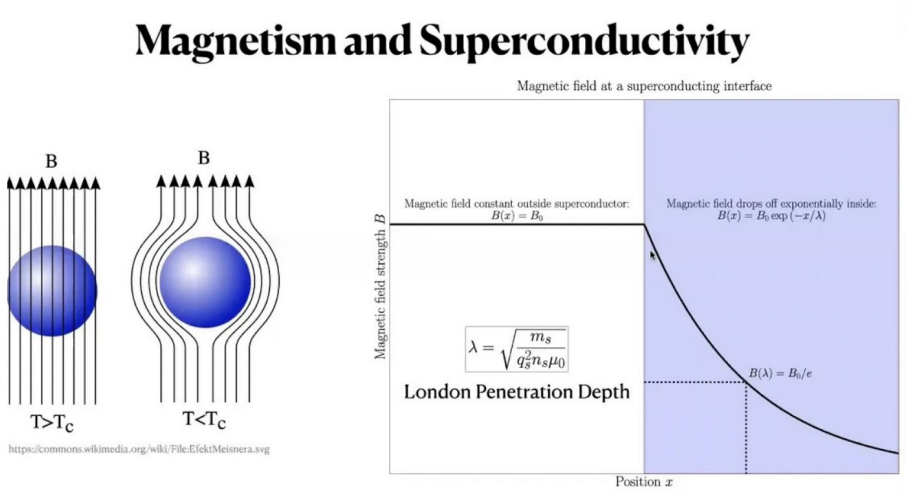






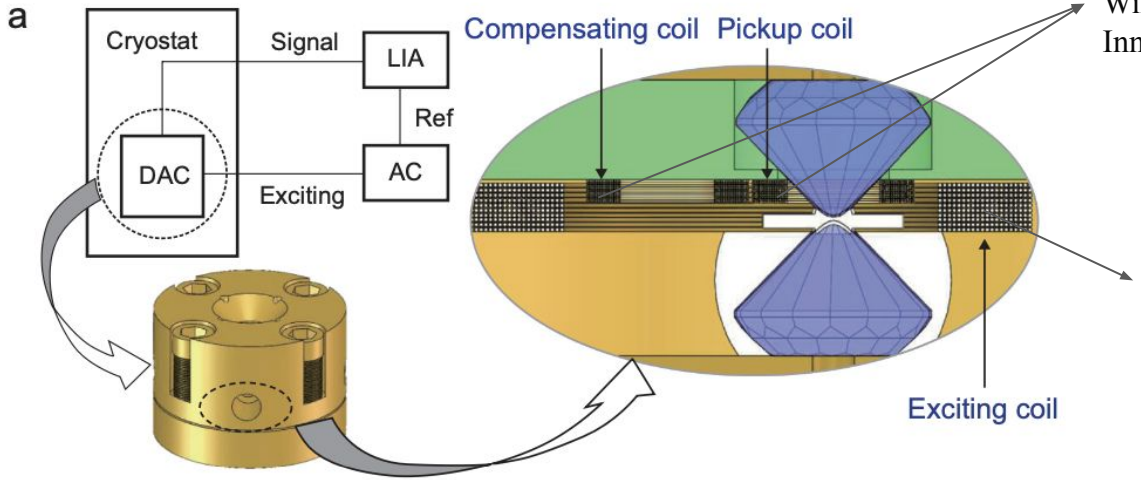
Identification of the superconducting state – Meissner effect

- **Below the critical temperature Tc**, a superconductor in a weak external magnetic field allows **the field to penetrate only slightly**, decaying to 1/e of its surface value over the London penetration depth.
- The superconducting transition is identified by monitoring **changes in magnetic susceptibility**, which reflect the emergence of **diamagnetism**.



Setup for AC magnetic susceptibility measurement

- **Compensating coil: oppositely connected to the pickup coil** to **cancel the constant susceptibility signal** originating from the surrounding metallic components.
- **LIA:** retrieve the amplitude and phase of small signals embedded in noise

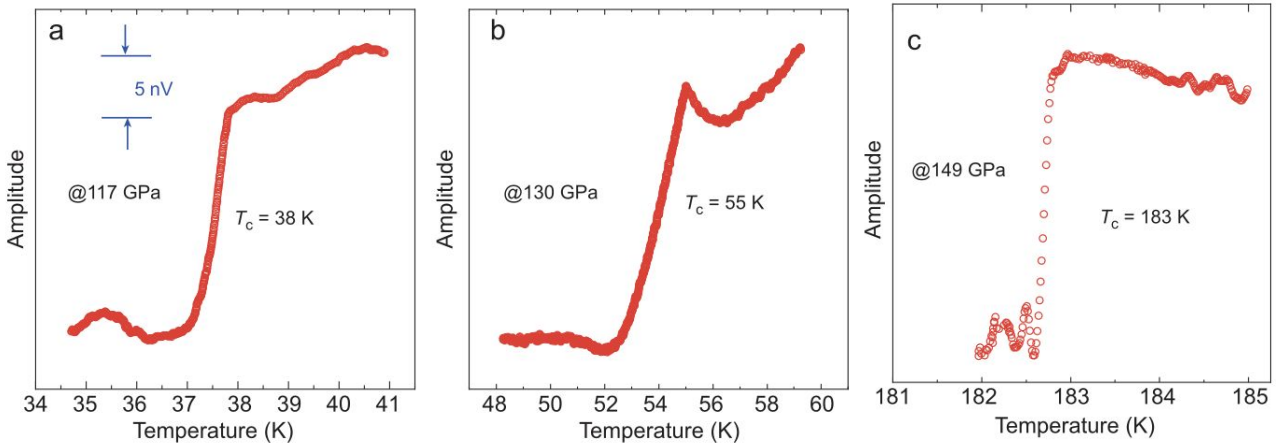


Number of turns: 180
Wire diameter: 20 μm (insulated copper wire)
Inner diameter: 2 mm

Number of turns: 300
Wire diameter: 38 μm (insulated copper wire)
Inner diameter: 9.4 mm
RMS current: 10 mA
Magnetic field: 0.75×10^{-4} T

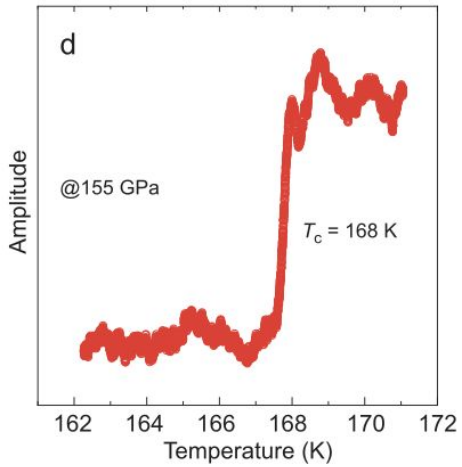
Sample preparation and first experiment

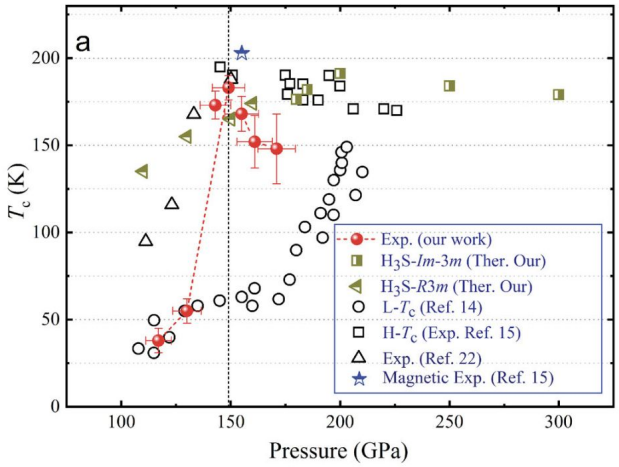
- culet size: 100 μm, preindented gasket: 20 μm, hole diameter: 60 μm
- In an argon glove box, H_2S was condensed into the gasket hole cooled in liquid nitrogen, and **the sample was subsequently compressed to at least 100 GPa at T < 100 K** to avoid possible decomposition.
- In the first run of the experiment, **superconducting transitions** were observed at **117 GPa and 130 GPa**, with **Tc values of 38 K and 55 K**, respectively.
- At 149 GPa, the sample was held at 300 K for seven days to allow the formation of the **H_3S phase**, resulting in a **Tc of 183 K**.



Second experiment and comparison

- At 155 GPa, Tc=168 K; further compression lowers Tc
- **Above 150 GPa**, the diameter of the diamond anvil culets is reduced, leading to a **decreased signal-to-noise ratio**.
- **Below 140 GPa**, **multiple hydrogen-rich H–S phases coexist** in the sample in addition to H_3S, and the low-Tc data can be reproduced by theoretical calculations based on stoichiometry.





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1. Xiaoli Huang, Xin Wang, Defang Duan, Bertil Sundqvist, Xin Li, Yanping Huang, Hongyu Yu, Fangfei Li, Qiang Zhou, Bingbing Liu, Tian Cui, High-temperature superconductivity in sulfur hydride evidenced by alternating-current magnetic susceptibility, *National Science Review*, Volume 6, Issue 4, July 2019, Pages 713–718, https://doi.org/10.1093/nsr/nwz061
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